BrightLearn Data Analytics Advanced SQL

Exercise 5 : NULL Functions

Table 1 : Employees

| employee-id | name | department | salary |
|---|---|---|---|
| 1 | Alice | HR | 5000 |
| 2 | Bob | IT | NULL |
| 3 | Charlie | NULL | 7000 |
| 4 | Dana | Finance | NULL |

Q1 Show all employees with their salary. If salary is NULL, display 0

Expected Columns: employee-id, name, salary_with_default

Code:

```
SELECT employee_id, name,
    COALESCE (Salary, 0) AS salary_with_default
FROM Employees;
```

Output table:

| employee_id | name | salary_with_default |
|---|---|---|
| 1 | Alice | 5000 |
| 2 | Bob | 0 |
| 3 | Charlie | 7000 |
| 4 | Dana | 0 |

Q2 Show all employee names with their department. If department is NULL, show "Not Assigned".

Expected columns: employee-id, name, department_name

code:

```
SELECT employee_id,
       name,
   COALESCE (department, 'NOT ASSIGNED') AS department_name
FROM employees;
```

output table

| employee_id | name | department_name |
|---|---|---|
| 1 | Alice | HR |
| 2 | Bob | IT |
| 3 | Charlie | Not Assigned |
| 4 | Dana | finance |

Table 2: orders

| order_id | customer_id | delivery_date |
|---|---|---|
| 101 | 201 | 2024-12-01 |
| 102 | 202 | NULL |
| 103 | NULL | 2024-12-03 |

Q3. Find orders with NULL customer_id using IS NULL().
Expected columns: order_id, customer_id

code:

```
SELECT order_id, customer
FROM orders
WHERE customer_id IS NULL;
```

output table

| order_id | customer_id |
|---|---|
| 103 | NULL |

Q4.

Show all orders. If delivery_date is NULL, show 'Pending'.

Expected columns: order_id, customer_id, delivery_stat

Code:

```
SELECT order_id, customer_id,
        COALESCE (CAST (delivery_date AS VARCHAR), 'Pending'
        AS delivery_status
FROM Orders;
```

Output table:

| Order_id | customer_id | delivery_status |
|---|---|---|
| 101 | 201 | 2024-12-01 |
| 102 | 202 | Pending |
| 103 | NULL | 2024-12-03 |

Q5

Table 3: Students.

| Student_id | name | grade |
|---|---|---|
| 1 | Ethan | 85 |
| 2 | Maya | NULL |
| 3 | Olivia | 90 |

Q5: Show all students and their grades. Replace NUL with 0.

Expected columns: student_id, name, final_grade.

Code:

```
SELECT student_id, name, COALESCE(grade, 0) AS final_grad
FROM Students.
```

output table for ques 5:

| student_id | name | final grade |
|---|---|---|
| 1 | Ethan | 85 |
| 2 | Maya | 0 |
| 3 | Olivia | 90 |

Q6: Count students who haven't been graded.
Expected Columns: not_graded count

code:

```
SELECT COUNT(*) AS not_graded_count
FROM students
WHERE grade IS NULL;
```

output table:

| not_graded_count |
|---|
| 1 |

Table 4: Products

| Product_id | name | Price | discount |
|---|---|---|---|
| 501 | keyboard | 25 | NULL |
| 502 | Mouse | 15 | 5 |
| 503 | Monitor | 100 | NULL |

Q7: Show product name, price, and final price after discount (assume 0 if discount is NULL).
Expected colums: product_id, name, final_price

code:

```
SELECT product_id, name,
       price - COALESCE(discount, 0) AS final_price
FROM Products;
```

output table:

| product_id | name | final_price |
|---|---|---|
| 501 | Keyboard | 25 |
| 502 | Mouse | 10 |
| 503 | Monitor | 100 |

Table 5: Customers

| customer_id | name | email |
|---|---|---|
| 1 | Linda | NULL |
| 2 | Joseph | joseph@mail.com |
| 3 | Nia | NULL |

Q8: Count how many customers have no email.
Expected columns: missing_email_count

code:

```
SELECT COUNT(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;
```

output table

| missing_email_count |
|---|
| 2 |

Q 9.
Show all custumers with email. If NULL, display "No Email".
Expected Columns: Customer-id, name, email_display

Code:

```
SELECT customer_id, name,
    COALESCE (email, 'No Email') AS email-display
FROM customers
```

output table:

| customer_id | name | email_display |
|---|---|---|
| 1 | Linda | No Email |
| 2 | Joseph | joseph@mail.com |
| 3 | Nia | No Email |

Table 6 : payments.

| Payment_id | method | status |
|---|---|---|
| 301 | Credit | NULL |
| 302 | Paypal | Success |
| 303 | NULL | Failed |

Ques 10: Show payment details with method replaced by "Unknown" if NULL.
Expected Columns: payment-id, method_display, status
Code:

```
SELECT payment_id,
    COALESCE (method, 'unknown') AS method_display, status
FROM payments;
```

output table:

| Payment_id | method_display | status |
|---|---|---|
| 301 | Credit | NULL |
| 302 | PayPal | Success |
| 303 | Unknown | failed |

Table 7: Inventory

| item_id | item_name | quantity |
|---|---|---|
| 1 | Pen | NULL |
| 2 | Notebook | 150 |
| 3 | Eraser | NULL |

Q11: Show items and their quantity (0 if NULL)
expected columns: item_id, item_name, quantity_checked

Code:

```
SELECT item_id, item_name,
        COALESCE (quantity, 0) AS quantity_checked
FROM Inventory;
```

output table:

| item_id | item_name | quantity checked |
|---|---|---|
| 1 | Pen | 0 |
| 2 | Notebook | 150 |
| 3 | Eraser | 0 |

Table 8: Employees_Extra

| emp_id | bonus | commission |
|---|---|---|
| 1 | NULL | 300 |
| 2 | 100 | NULL |
| 3 | NULL | NULL |

Q12: Show employee ID and the first available value among bonus or commission.
Expected Colums: emp_id, first_available_reward

Code:

```
SELECT emp_id,
    COALESCE (bonus, commission) AS first available_rewar
FROM Employee_Extra;
```

output table:

| emp_id | first_available_reward |
|---|---|
| 1 | 300 |
| 2 | 100 |
| 3 | NULL |

Table 9: Classes

| class_id | subject | room |
|---|---|---|
| 11 | Math | NULL |
| 12 | Science | LabA |
| 13 | English | NULL |

Q13. Count classes that dont have a room assigned
Expected Columns: no_room_count

Code for Q13

```
SELECT COUNT(*) AS no_room_count
FROM Classes
WHERE room IS NULL;
```

output table:

| no_room_count |
|---|
| 2. |

Table 10: Attendance.

| student_id | date | Status |
|---|---|---|
| 1 | 2025-04-01 | NULL |
| 2 | 2025-04-01 | Present |
| 3 | 2025-04-01 | Absent |

Q14: Show attendance records with status. Replace NULL with 'Not Marked'.

Expected columns: student_id, date, attendance_status.

Code:

```
SELECT student_id, date,
    COALESCE(status, 'Not Marked') AS attendance_status
FROM Attendance
```

↓

outcome table:

| student_id | date | attendance_status |
|---|---|---|
| 1 | 2025-04-01 | Not Marked |
| 2 | 2025-04-01 | Present |
| 3 | 2025-04-01 | Absent |

Table 11 : Bank_Accounts

| account_id | account_type | balance |
|---|---|---|
| A1 | Savings | NULL |
| A2 | Current | 5000 |
| A3 | NULL | 2000 |

Q15: Show account ID, account_type (or 'Unknown'), and balance (or 0).

Expected Columns: account_id, type_display, balance_chec

code:

```
SELECT account_id,
    COALESCE (account_type, 'unknown') AS type_display,
    COALESCE (balance, 0) AS balance_checked
FROM Bank_Accounts;
```

Output Table:

| account_id | type_display | balance_checked |
|---|---|---|
| A1 | Savings | 0 |
| A2 | Current | 5000 |
| A | Unknown | 2000 |

Table 12 : Projects

| project_id | title | start_date | end_date |
|---|---|---|---|
| 1 | Website Revamp | 2025-01-10 | NULL |
| 2 | Mobile App | NULL | 2025-06-01 |
| 3 | Data Migration | NULL | NULL |

Q16. Show all projects with a start date. If start_date is NULL, display 'TBD'.

Expected Columns: project_id, title, start_display

Code:

```
SELECT project_id, title,
        COALESCE(CAST(start_date AS VARCHAR), 'TBD') AS
start_display
FROM projects;
```

Output table:

| project_id | title | start_display |
|---|---|---|
| 1 | Website Revamp | 2025-01-10 |
| 2 | Mobile App | TBD |
| 3 | Data Migration | TBD |

Table 13: Reviews

| review_id | product_id | comment | rating |
|---|---|---|---|
| 1 | 501 | Great product | 4 |
| 2 | 502 | NULL | NULL |
| 3 | 503 | Work fine | 3 |

Q17. Display reviews showing comment (or 'NO comment') and rating (or 0).
Expected columns: review_id, product_id, comment_display, rating_display

code:

```
SELECT review_id, product_id,
    COALESCE (comment, 'NO comment') AS comment_display,
    COALESCE (rating, 0) AS rating_display
FROM Reviews;
```

output table:

| review_id | product_id | comment_display | rating_display |
|---|---|---|---|
| 1 | 501 | Great product | 4 |
| 2 | 502 | NO comment | 0 |
| 3 | 503 | Works fine | 3 |

Table 14: Suppliers.

| supplier_id | name | phone | alt_phone |
|---|---|---|---|
| 1 | Global Goods | NULL | 123456789 |
| 2 | Best Supplies | 987654321 | NULL |
| 3 | Valuesource | NULL | NULL |

Q18: Show the supplier contact number. Use COALESCE (phone, alt_phone, 'NO contact').
Expected columns: supplier_id, name, contact_number

code for Q18

```
SELECT supplier_id, name,
    COALESCE(phone, alt_phone, 'No Contact') AS Contact_num
FROM suppliers
```

Output table,

| supplier_id | name | contact_number |
|---|---|---|
| 1 | Global Goods | 123456789 |
| 2 | Best supplies | 987654321 |
| 3 | valuesource | 'No Contact' |

Table 15: use_settings

| user_id | theme | language | timezone |
|---|---|---|---|
| 1 | NULL | English | NULL |
| 2 | Dark | NULL | UTC+1 |
| 3 | NULL | NULL | NULL |

Q19. Show all users and their preferences. Replace all NULL with defaults:

- Theme → "Light"
- Language → "English"
- Timezone → "UTC"

Expected Columns: user_id, theme_set, language_set, timezone_s

code:

```
V'
```

Code for 19:

```
SELECT user_id,
    COALESCE (theme, 'Light') AS theme_set,
    COALESCE (language, 'English') AS language_set,
    COALESCE (timezone, 'UTC') AS timezone_set
FROM user_settings
```

output table:

| user_id | theme_set | language_set | timezone_set |
|---|---|---|---|
| 1 | Light | English | UTC |
| 2 | Dark | English | UTC+1 |
| 3 | Light | English | UTC |

Table 16: Maintenance

| record_id | machine_id | issue | technician |
|---|---|---|---|
| 1 | M101 | Overheating | NULL |
| 2 | M102 | NULL | NULL |
| 3 | M103 | Jammed | Alex |

Q20. show maintenance log with:
- issue → default to 'Unknown Issue'
- technician → default to "Not Assigned"

Expected columns: record_id, machine_id, issue_log, technician_name

code

```
SELECT record_id, machine_id,
    COALESCE (issue, 'Unknown Issue') AS issue_log
    COALESCE (technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

output table:

| record_id | machine_id | issue_log | technian_name |
|---|---|---|---|
| 1 | M101 | Overheating | NA Assigned |
| 2 | M102 | Unknown Issue | NA Assigned |
| 3 | M103 | Jammed | Alex |